hence $A^n = \bigoplus_{i=1}^r nd_iV_i$ as a representation of A. Thus by Lemma 3.1.6, $X = \bigoplus_{i=1}^r m_iV_i$, as desired.

3.4. Filtrations

Let A be an algebra. Let V be a representation of A.

Definition 3.4.1. A (finite) **filtration** of V is a sequence of subrepresentations $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$.

Lemma 3.4.2. *Any finite dimensional representation V of an algebra A admits a finite filtration $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ such that the successive quotients V_i/V_{i-1} are irreducible.*

Proof. The proof is by induction in $\dim(V)$. The base is clear, and only the induction step needs to be justified. Pick an irreducible subrepresentation $V_1 \subset V$, and consider the representation $U = V/V_1$. Then by the induction assumption U has a filtration $0 = U_0 \subset U_1 \subset \cdots \subset U_{n-1} = U$ such that U_i/U_{i-1} are irreducible. Define V_i for $i \geq 2$ to be the preimages of U_{i-1} under the tautological projection $V \to V/V_1 = U$. Then $0 = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_n = V$ is a filtration of V with the desired property. □

3.5. Finite dimensional algebras

Definition 3.5.1. The **radical** of a finite dimensional algebra A is the set of all elements of A which act by 0 in all irreducible representations of A. It is denoted $\mathrm{Rad}(A)$.

Proposition 3.5.2. $\mathrm{Rad}(A)$ *is a two-sided ideal.*

Proof. Easy. □

Proposition 3.5.3. *Let A be a finite dimensional algebra.*

(i) Let I be a nilpotent two-sided ideal in A; i.e., $I^n = 0$ for some n. Then $I \subset \mathrm{Rad}(A)$.

(ii) $\mathrm{Rad}(A)$ is a nilpotent ideal. Thus, $\mathrm{Rad}(A)$ is the largest nilpotent two-sided ideal in A.